Codekata Report:

Name: GOKUL

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Specialization: Cyber Security

Completion Year: 2028

Section: Section-A

1. Problem Statement:Write a code to get the input in the given format and print the output in the given format.

Input Description:First-line indicates two integers which are the size of array and 'K' value.Second-line indicates an integer contains elements of an array.

Output Description:Print the taken input in the same format.

Sample Input:5 31 2 3 4 5

Sample Output:5 31 2 3 4 5

Explanation:self

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Input Description:

First-line indicates two integers which are the size of array and 'K' value.
Second-line indicates an integer contains elements of an array.

Output Description:

Print the taken input in the same format.

Completion Status: Not Completed

Concepts Included:

Input/Output

Language Used: PYTHON 3

Source Code:

```
#A Simple Hello World
```

```
print("Hello World")

#Getting input via STDIN
userInput = input()
print("The Input Provided is: " + userInput)
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

Hello World
The Input Provided is: 5 3

Compilation Status: Failed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

Hello World
The Input Provided is: 4 2

Compilation Status: Failed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

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Expected Output:

< hidden >

Output:

Hello World
The Input Provided is: 5 4

Compilation Status: Failed

Execution Time:

0.014s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

Hello World
The Input Provided is: 3 1

Compilation Status: Failed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

Hello World
The Input Provided is: 1 1

Compilation Status: Failed

Execution Time:

0.011s

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2. Problem Statement:Given 2 numbers N and M add both the numbers and check whether the sum is odd or even.

Input Description:The input consists of two integers, N and M.

Output Description:The output is 'odd' if the sum of N and M is odd, and 'even' if the sum is even.

Sample Input:9 2

Sample Output:odd

Completion Status: Completed

Concepts Included:

basics

mathematics

Language Used: PYTHON 3

Source Code:

```
# Read two integers from input
N, M = map(int, input().split())

# Calculate sum
total = N + M

# Check odd or even
if total % 2 == 0:
print("even")
else:
print("odd")
```

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Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

odd

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

even

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

even

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

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odd

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

odd

Compilation Status: Passed

Execution Time:

0.01s

GOKUL (gokulnarendran46@gmail.com)

3. Problem Statement:Given a number N, check whether it is prime or not. Print 'yes' if it is prime else print 'no'.

Input Description:The input consists of a single integer N.

Output Description:The output is 'yes' if N is prime, otherwise 'no'.

Sample Input:123

Sample Output:no

Completion Status: Completed

Concepts Included:

mathematics

Language Used: PYTHON 3

Source Code:

```
N = int(input())

if N <= 1:
```

```
print("no")
else:
is_prime = True
for i in range(2, int(N**0.5) + 1):
if N % i == 0:
is_prime = False
break

if is_prime:
print("yes")
else:
print("no")
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.01s

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TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.011s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

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0.01s

4. Problem Statement:Given a number N, print yes if the number is multiple of 7 else print no.

Input Description:The input consists of a single integer N.

Output Description:Print 'yes' if N is a multiple of 7, otherwise print 'no'.

Sample Input:49

Sample Output:yes

Completion Status: Completed

Concepts Included:

mathematics

Language Used: PYTHON 3

Source Code:

```
N = int(input())

if N % 7 == 0:
print("yes")
else:
print("no")
```

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Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.013s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.014s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

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0.014s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.013s

5. Question:Given a number N, find its next immediate greater power of 2(i.e 2^1, 2^2, 2^3...).

Input Description:The input consists of a number N where N <= 1000.

Sample Input:4

Sample Output:8

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Completion Status: Completed

Concepts Included:

mathematics

Language Used: PYTHON 3

Source Code:

```
N = int(input())

power = 1
while power <= N:
power *= 2

print(power)
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

32

Compilation Status: Passed

Execution Time:

0.011s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

128

Compilation Status: Passed

Execution Time:

0.014s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

2

Compilation Status: Passed

Execution Time:

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0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

4

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

512

Compilation Status: Passed

Execution Time:

0.01s

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6. Problem Statement:Given a string S, print 'yes' if it is a palindrome or 'no' if it is not a palindrome.

Sample Input:lappal

Sample Output:yes

Completion Status: Completed

Concepts Included:

strings

Language Used: PYTHON 3

Source Code:

```
s = input()

if s == s[::-1]:
print("yes")
else:
print("no")
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.011s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.01s

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TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

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0.01s

7. Question:Given a string S, print 'yes' if it has a vowel in it else print 'no'.

Sample Input:codekata

Sample Output:yes

Completion Status: Completed

Concepts Included:

strings

Language Used: PYTHON 3

Source Code:

```
s = input().lower()

vowels = "aeiou"
found = False

for ch in s:
if ch in vowels:
found = True
break

if found:
print("yes")
else:
print("no")
```

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Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.014s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.012s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

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yes

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.014s

8. Question:Given 3 numbers N , L and R. Print 'yes' if N is between L and R else print 'no'.

Sample Input:32 6

Sample Output:yes

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Completion Status: Completed

Concepts Included:

mathematics

basics

Language Used: PYTHON 3

Source Code:

```
N = int(input())
L, R = map(int, input().split())

if L < N < R:
print("yes")
```

else:
print("no")

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.013s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.013s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

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no

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.014s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.014s

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9. Question:Given a floating point number with 1 decimal place round it off to nearest greater integer and print it.

Sample Input:2.6

Sample Output:3

Completion Status: Completed

Concepts Included:

mathematics

Language Used: PYTHON 3

Source Code:

```
import math

n = float(input())
print(math.ceil(n))
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

2

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

22

Compilation Status: Passed

Execution Time:

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0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

0

Compilation Status: Passed

Execution Time:

0.011s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

4

Compilation Status: Passed

Execution Time:

0.015s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

3

Compilation Status: Passed

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Execution Time:

0.015s

10. Question:Given a string S, print it after changing the middle element to * (if the length of the string is even, change the 2 middle elements to *).

Sample Input:hello

Sample Output:he*lo

Completion Status: Completed

Concepts Included:

array

strings

Language Used: PYTHON 3

Source Code:

```
s = input()
n = len(s)

if n % 2 == 1:
mid = n // 2
result = s[:mid] + '*' + s[mid+1:]
else:
mid = n // 2
result = s[:mid-1] + '**' + s[mid+1:]

print(result)
```

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Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

*

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

sa**ad

Compilation Status: Passed

Execution Time:

0.014s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1*3

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

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Output:

du**ng

Compilation Status: Passed

Execution Time:

0.014s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

he*lo

Compilation Status: Passed

Execution Time:

0.014s

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11. Problem Statement:Given a number N, print 'yes' if it is composite else print 'no'.

Sample Input:123

Sample Output:yes

Completion Status: Completed

Concepts Included:

mathematics

basics

Language Used: PYTHON 3

Source Code:

```
N = int(input())

if N <= 1:
```

```
print("no")   # 0, 1 are not composite
else:
composite = False
for i in range(2, int(N**0.5) + 1):
if N % i == 0:
composite = True
break

if composite:
print("yes")
else:
print("no")
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.01s

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TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

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0.01s

12. Problem Statement:Given a number N, print its factors.

Input Description:Input Size : n<=1000

Sample Input:6

Sample Output:1 2 3 6

Completion Status: Completed

Concepts Included:

mathematics

Language Used: PYTHON 3

Source Code:

```
N = int(input())
factors = []

# Check only up to sqrt(N)
for i in range(1, int(N**0.5) + 1):
if N % i == 0:
factors.append(i)
if i != N // i:
factors.append(N // i)

# Sort factors and print
factors.sort()
print(*factors)
```

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Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1 2 5 10

Compilation Status: Passed

Execution Time:

0.013s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1 2 3 4 6 12

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1 2 4 8

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1 3 9

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Compilation Status: Passed

Execution Time:

0.014s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1 2 3 6

Compilation Status: Passed

Execution Time:

0.01s

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13. Problem Statement:Given a number N, print 'yes' if it is a multiple of 13 else print 'no'.

Sample Input:26

Sample Output:yes

Completion Status: Completed

Concepts Included:

mathematics

Language Used: PYTHON 3

Source Code:

```
N = int(input())

if N % 13 == 0:
print("yes")
else:
print("no")
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.011s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

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0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.01s

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14. Problem Statement:Given 2 numbers N,M. Print 'yes' if their product is a perfect square else print 'no'.

Sample Input:5 5

Sample Output:yes

Completion Status: Completed

Concepts Included:

mathematics

basics

Language Used: PYTHON 3

Source Code:

```
import math

N, M = map(int, input().split())

product = N * M

# Find the integer square root
sqrt_product = int(math.sqrt(product))

# Check if the square of the integer root equals the product
if sqrt_product * sqrt_product == product:
print("yes")
else:
print("no")
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

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Output:

no

Compilation Status: Passed

Execution Time:

0.011s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.011s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.011s

TestCase5:

Input:

< hidden >

Expected Output:

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< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.01s

15. Problem Statement:Given a number N, print the odd digits in the number(space seperated) or print -1 if there is no odd digit in the given number.

Input Description:The input consists of a single integer N, where N <= 100000.

Output Description:The output should be the odd digits of N, space-separated, or -1 if no odd digits are present.

Sample Input:2143

Sample Output:1 3

Completion Status: Completed

Concepts Included:

array

mathematics

Language Used: PYTHON 3

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Source Code:

```
N = input()  # Read the number as a string
odd_digits = []

for digit in N:
if int(digit) % 2 == 1:
odd_digits.append(digit)

if odd_digits:
print(" ".join(odd_digits))
else:
print(-1)
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

0.011s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1 3

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1 1 1 1 1

Compilation Status: Passed

Execution Time:

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0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.015s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

5

Compilation Status: Passed

Execution Time:

0.015s

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16. Problem Statement:Kabali is a brave warrior who with his group of young ninjas moves from one place to another to fight against his opponents. Before Fighting he just calculates one thing, the difference between his ninja number and the opponent's ninja number. From this difference he decides whether to fight or not. Kabali's ninja number is never greater than his opponent.

Input Description:The input contains two numbers in every line. These two numbers in each line denotes the number ninjas in Kabali's clan and his opponent's clan .

Output Description:print the absolute difference of number of ninjas between Kabali's clan and his opponent's clan. Each output should be in seperate line.

Sample Input:100 200

Sample Output:100

Completion Status: Completed

Concepts Included:

mathematics

basics

Language Used: PYTHON 3

Source Code:

```
# Read input
kabali, opponent = map(int, input().split())

# Calculate absolute difference
difference = abs(opponent - kabali)

# Print the difference
print(difference)
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

90

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

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Expected Output:

< hidden >

Output:

88

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

900

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

3

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

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< hidden >

Expected Output:

< hidden >

Output:

1023

Compilation Status: Passed

Execution Time:

0.01s

17. Problem Statement:Given length L and breadth B of a farm, print the area of the farm upto 5 decimal decimals.

Sample Input:1.626 2.31

Sample Output:3.75606

Completion Status: Not Completed

Concepts Included:

mathematics

Language Used: PYTHON 3

Source Code:

```
# Read input values
L, B = map(float, input().split())

# Calculate area
area = L * B

# Print result with exactly 5 decimal places using format()
print("{:.5f}".format(area))
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

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< hidden >

Output:

10.00000

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

17.20577

Compilation Status: Failed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

81.00000

Compilation Status: Passed

Execution Time:

0.009s

TestCase4:

Input:

< hidden >

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Expected Output:

< hidden >

Output:

0.00000

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

640.00000

Compilation Status: Passed

Execution Time:

0.01s

GOKUL (gokulmahendran46@gmail.com)

18. Problem Statement:Given a string S, print 2 strings such that first string containing all characters in odd position(s) and other containing all characters in even position(s).

Sample Input:XCODE

Sample Output:XOE CD

Completion Status: Completed

Concepts Included:

strings

array

Language Used: PYTHON 3

Source Code:

```
# Read input
S = input().strip()

# Odd and even position characters
odd = ""
even = ""

for i in range(len(S)):
if i % 2 == 0:     # index 0 → position 1 (odd)
odd += S[i]
else:
even += S[i]

# Print result
print(odd, even)
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

Snya undy

Compilation Status: Passed

Execution Time:

0.014s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

246 35

Compilation Status: Passed

GOKUL (gokulmahendran46@gmail.com)

Execution Time:

0.014s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

gv ui

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

Sun Le

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

Bbbakhe aalcsep

GOKUL (gokulmahendran46@gmail.com)

Compilation Status: Passed

Execution Time:

0.011s

19. Problem Statement:Given 2 numbers N,M find the GCD of N and M.If it cannot be found for given number(s) then print -1

Sample Input:10 5

Sample Output:5

Completion Status: Not Completed

Concepts Included:

mathematics

basics

Language Used: PYTHON 3

Source Code:

```
import math
N, M = map(int, input().split())
if N == 0 and M == 0:
print(-1)
else:
print(math.gcd(N, M));
```

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

12

Compilation Status: Failed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.014s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

2

Compilation Status: Passed

GOKUL (gokulmahendran46@gmail.com)

Execution Time:

0.014s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

3

Compilation Status: Passed

Execution Time:

0.013s

GOKUL (gokulmohandran4@gmail.com)

20. Problem Statement:Write a program to calculate the total surface area and volume of cuboid.

Input Description:Input contains three space separated positive integers L, B, H denoting the length, width and height of cuboid respectively.

Output Description:The output should be the total surface area and volume of the cuboid, separated by a space.

Sample Input:1 2 3

Sample Output:22 6

Completion Status: Completed

Concepts Included:

mathematics

Language Used: PYTHON 3

Source Code:

```
L, B, H = map(int, input().split())

# Total Surface Area
tsa = 2 * (L*B + B*H + H*L)
```

```
# Volume
vol = L * B * H

print(tsa, vol)
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

10 2

Compilation Status: Passed

Execution Time:

0.015s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

52 24

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

GOKUL (gokulmahendran46@gmail.com)

Output:

24 8

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

6 1

Compilation Status: Passed

Execution Time:

0.014s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

22 6

Compilation Status: Passed

Execution Time:

0.009s

GOKUL (gokulmahendran46@gmail.com)

21. Problem Statement:Given a number N, print its reverse.

Input Description:Input Size : n <= 1000

Sample Input:10

Sample Output:1

Completion Status: Completed

Concepts Included:

mathematics

Language Used: PYTHON 3

Source Code:

```
N = int(input())
rev = 0

while N > 0:
digit = N % 10
rev = rev * 10 + digit
N //= 10

print(rev)
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

62

Compilation Status: Passed

Execution Time:

0.013s

TestCase2:

Input:

< hidden >

Expected Output:

GOKUL (gokulmahendran46@gmail.com)

< hidden >

Output:

87618

Compilation Status: Passed

Execution Time:

0.012s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

19

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

21

Compilation Status: Passed

Execution Time:

0.013s

TestCase5:

Input:

< hidden >

GOKUL (gokulmahendran46@gmail.com)

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.01s

22. Problem Statement:Given a number N followed by N elements which can be arranged in ascending order with maximum one element update. Print the index of the element which has to be changed else print '-1' if the updation not neccassary or if the given input needs more than one update to form ascending order.

Sample Input:71 2 4 3 5 6 851 10 3 14 5

Sample Output:2-1

Completion Status: Completed

Concepts Included:

mathematics

Language Used: PYTHON 3

GOKUL (gokulmahendran46@gmail.com)

Source Code:

```
def find_update_index(arr):
n = len(arr)
count = 0
idx = -1

for i in range(n - 1):
if arr[i] > arr[i + 1]:
count += 1
idx = i
if count > 1:
return -1

if count == 0:
return -1

# Check if updating arr[idx] or arr[idx+1] fixes the array
```

```
temp1 = arr[:]
temp1[idx] = temp1[idx + 1]  # try updating current element
if all(temp1[i] <= temp1[i + 1] for i in range(n - 1)):
return idx

temp2 = arr[:]
temp2[idx + 1] = temp2[idx]  # try updating next element
if all(temp2[i] <= temp2[i + 1] for i in range(n - 1)):
return idx + 1

return -1

# Read input
N = int(input())
arr = list(map(int, input().split()))
print(find_update_index(arr))
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

GOKUL (gokulmahendran46@gmail.com)

Execution Time:

0.015s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

2

Compilation Status: Passed

Execution Time:

0.014s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

0

GOKUL (gokulmahendran46@gmail.com)

Compilation Status: Passed

Execution Time:

0.01s

23. Problem Statement:Given 3 numbers a,b,c print a*b mod c.

Sample Input:5 3 2

Sample Output:1

Completion Status: Completed

Concepts Included:

mathematics

Language Used: PYTHON 3

Source Code:

```
a, b, c = map(int, input().split())
result = (a * b) % c
print(result)
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

0

Compilation Status: Passed

Execution Time:

0.014s

TestCase2:

Input:

GOKUL (gokulmahendran46@gmail.com)

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.014s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

4

Compilation Status: Passed

Execution Time:

0.013s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

3

Compilation Status: Passed

Execution Time:

0.014s

TestCase5:

GOKUL (gokulmahendran46@gmail.com)

Input:

< hidden >

Expected Output:

< hidden >

Output:

0

Compilation Status: Passed

Execution Time:

0.013s

24. Problem Statement:Given a number N, print the product of the digits.

Input Description:Input Size : N <= 100000000000

Sample Input:2143

Sample Output:24

Completion Status: Completed

Concepts Included:

mathematics

Language Used: PYTHON 3

GOKUL (gokulmahendran46@gmail.com)

Source Code:

```
N = input().strip()  # Read as string to handle large numbers
product = 1

for digit in N:
product *= int(digit)

print(product)
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

0

Compilation Status: Passed

Execution Time:

0.009s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

256

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.014s

TestCase4:

GOKUL (gokulmahendran46@gmail.com)

Input:

< hidden >

Expected Output:

< hidden >

Output:

24

Compilation Status: Passed

Execution Time:

0.015s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

0

Compilation Status: Passed

Execution Time:

0.01s

GOKUL (gokulmahendran46@gmail.com)

25. Problem Statement:Given 3 numbers A,B,C find the sum of Arithmetic Series with a=A, d=B and n=C

Sample Input:1 1 2

Sample Output:3

Completion Status: Completed

Concepts Included:

mathematics

Language Used: PYTHON 3

Source Code:

```
A, B, C = map(int, input().split())

# Arithmetic series sum formula
S = C * (2 * A + (C - 1) * B) // 2
print(S)
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.015s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

84

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

GOKUL (gokulmahendran46@gmail.com)

Expected Output:

< hidden >

Output:

108

Compilation Status: Passed

Execution Time:

0.013s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

243

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

13

Compilation Status: Passed

Execution Time:

0.014s

GOKUL (gokulmahendran46@gmail.com)

26. Problem Statement:Find the minimum among 10 numbers.

Input Description:The input consists of 10 space-separated integers.

Output Description:The output is the minimum of the given 10 integers.

Sample Input:5 4 3 2 1 7 6 10 8 9

Sample Output:1

Completion Status: Completed

Concepts Included:

basics

mathematics

Language Used: PYTHON 3

Source Code:

```
numbers = list(map(int, input().split()))
print(min(numbers))
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

0

Compilation Status: Passed

Execution Time:

0.014s

TestCase2:

Input:

< hidden >

Expected Output:

GOKUL (gokulmahendran46@gmail.com)

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.014s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.015s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-2

Compilation Status: Passed

Execution Time:

0.014s

TestCase5:

Input:

< hidden >

GOKUL (gokulmahendran46@gmail.com)

Expected Output:

< hidden >

Output:

3

Compilation Status: Passed

Execution Time:

0.01s

27. Problem Statement:Given 2 numbers N and K followed by elements of N .Print 'yes' if K exists else print 'no'.

Input Description:The input consists of two integers, N and K, followed by N integers.

Output Description:The output is 'yes' if K is found among the N integers, otherwise 'no'.

Sample Input:4 21 2 3 3

Sample Output:yes

GOKUL (gokulmahendran46@gmail.com)

Completion Status: Completed

Concepts Included:

basics

array

Language Used: PYTHON 3

Source Code:

```
# Read N and K
N, K = map(int, input().split())

# Read N numbers
arr = list(map(int, input().split()))

# Check if K exists
if K in arr:
print("yes")
else:
print("no")
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

GOKUL (gokulmahendran46@gmail.com)

0.014s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.014s

GOKUL (gokulmahendran46@gmail.com)

28. Problem Statement:Given 2 numbers N and K followed by N elements,print the number of repetition of K otherwise print '-1' if the element not found.

Sample Input:6 21 2 3 5 7 8

Sample Output:1

Completion Status: Completed

Concepts Included:

basics

mathematics

array

Language Used: PYTHON 3

Source Code:

```
# Read N and K
N, K = map(int, input().split())

# Read N elements
arr = list(map(int, input().split()))

# Count occurrences of K
count = arr.count(K)

# Print result
print(count if count > 0 else -1)
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

GOKUL (gokulmahendran46@gmail.com)

Output:

-1

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

3

Compilation Status: Passed

Execution Time:

0.015s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.014s

TestCase5:

Input:

< hidden >

Expected Output:

GOKUL (gokulmahendran46@gmail.com)

< hidden >

Output:

2

Compilation Status: Passed

Execution Time:

0.01s

29. Problem Statement:Count the number of digits of a given number N.Size of the integer ranges from 1<N<100000000

Sample Input:548

Sample Output:3

Completion Status: Completed

Concepts Included:

mathematics

Language Used: PYTHON 3

Source Code:

```
N = int(input())
print(len(str(N)))
```

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

3

Compilation Status: Passed

Execution Time:

0.014s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

7

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

4

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

GOKUL (gokulmahendran46@gmail.com)

Execution Time:

0.013s

30. Problem Statement:Given a string S, print it without using semicolon in your program.

Sample Input:hello world

Sample Output:hello world

Completion Status: Completed

Concepts Included:

strings

array

Language Used: PYTHON 3

Source Code:

```
S = input()
print(S)
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

hello world

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

GOKUL (gokulmahendran46@gmail.com)

< hidden >

Expected Output:

< hidden >

Output:

guvi geeks

Compilation Status: Passed

Execution Time:

0.013s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

123

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

4as

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

GOKUL (gokulmahendran46@gmail.com)

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.009s

31. Problem Statement:Given a number N, find the nearest greater multiple of 10.

Input Description:Input Size : N <= 10000

Sample Input:3

Sample Output:10

Completion Status: Completed

Concepts Included:

mathematics

Language Used: PYTHON 3

GOKUL (gokulmahendran46@gmail.com)

Source Code:

```
# Read input
N = int(input())

# Calculate nearest greater multiple of 10
nearest_multiple = ((N // 10) + 1) * 10

# Print the result
print(nearest_multiple)
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

100

Compilation Status: Passed

Execution Time:

0.017s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

10

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

10

Compilation Status: Passed

Execution Time:

0.015s

TestCase4:

GOKUL (gokulmahendran46@gmail.com)

Input:

< hidden >

Expected Output:

< hidden >

Output:

20

Compilation Status: Passed

Execution Time:

0.016s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

40

Compilation Status: Passed

Execution Time:

0.01s

GOKUL (gokulmahendran46@gmail.com)

32. Problem Statement:Given 2 numbers N,M. Find their difference and check whether it is even or odd.

Sample Input:5 5

Sample Output:even

Completion Status: Completed

Concepts Included:

mathematics

Language Used: PYTHON 3

Source Code:

```
# Read two integers from input (space-separated)
n, m = map(int, input().split())

# Compute the difference
diff = n - m

# Check even or odd using modulus operator
if diff % 2 == 0:
print("even")
else:
print("odd")
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

even

Compilation Status: Passed

Execution Time:

0.012s

GOKUL (gokulmahendran46@gmail.com)

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

odd

Compilation Status: Passed

Execution Time:

0.014s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

even

Compilation Status: Passed

Execution Time:

0.014s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

odd

Compilation Status: Passed

Execution Time:

0.012s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

even

Compilation Status: Passed

Execution Time:

GOKUL (gokulmahendran46@gmail.com)

0.014s

33. Problem Statement:Write a program to print the sum of the firs K natural numbers.

Input Description:Input Size : n <= 100000

Sample Input:3

Sample Output:6

Completion Status: Completed

Concepts Included:

basics

mathematics

Language Used: PYTHON 3

Source Code:

```
# Read K from input
k = int(input())

# Compute sum of first K natural numbers using formula
total = k * (k + 1) // 2

# Print the result
print(total)
```

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

210

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

6

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

820

Compilation Status: Passed

GOKUL (gokulmahendran46@gmail.com)

Execution Time:

0.015s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

15

Compilation Status: Passed

Execution Time:

0.01s

GOKUL (gokulmahendran24@gmail.com)

34. Problem Statement:Given a number N followed by N elements, find the second smallest element.If it cannot be found then print -1

Input Description:Input Size : N <= 100000 (ie do it in O(log n) time complexity)

Sample Input:51 2 3 4 5

Sample Output:2

Completion Status: Completed

Concepts Included:

array

Language Used: PYTHON 3

Source Code:

```
# Read input
N = int(input())
arr = list(map(int, input().split()))

if N < 2:
print(-1)
else:
first = second = float('inf')
for num in arr:
```

```
if num < first:
second = first
first = num
elif first < num < second:
second = num

# Check if second smallest exists
if second == float('inf'):
print(-1)
else:
print(second)
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-5

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

0.013s

TestCase3:

GOKUL (gokulmahendran46@gmail.com)

Input:

< hidden >

Expected Output:

< hidden >

Output:

5

Compilation Status: Passed

Execution Time:

0.014s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

3

Compilation Status: Passed

Execution Time:

0.014s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

3

Compilation Status: Passed

Execution Time:

0.015s

GOKUL (gokulmahendran46@gmail.com)

35. Problem Statement:Given 2 numbers N and K followed by N elements, find the Kth smallest element. If the element cannot be found then print -1

Input Description:The input consists of two numbers N and K, followed by N elements. N <= 100000.

Output Description:The output is the Kth smallest element. If the element cannot be found, print -1.

Sample Input:5 21 1 2 4 5

Sample Output:2

Completion Status: Completed

Concepts Included:

array

ABCO

Accolite

Amazon

Cisco

Hike

Microsoft

Snapdeal

VMWare

guvi-learning-path

GOKUL (gokulmahendran46@gmail.com)

Language Used: PYTHON 3

Source Code:

```
# Read input
N, K = map(int, input().split())
arr = list(map(int, input().split()))

# Remove duplicates to get distinct elements
arr = list(set(arr))

# Check if Kth smallest exists
if K > len(arr) or K <= 0:
print(-1)
else:
arr.sort()
```

```
print(arr[K-1])
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

5

Compilation Status: Passed

Execution Time:

0.014s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

0.013s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

GOKUL (gokulmahendran46@gmail.com)

3

Compilation Status: Passed

Execution Time:

0.015s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

2

Compilation Status: Passed

Execution Time:

0.014s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

4

Compilation Status: Passed

Execution Time:

0.01s

GOKUL (gokulmahendran46@gmail.com)

36. Problem Statement:Given a string S, find its length(including the spaces)without using any pre-defined functions.

Sample Input:codekata

Sample Output:8

Completion Status: Completed

Concepts Included:

strings

Language Used: PYTHON 3

Source Code:

```
# Read input
S = input()

# Initialize length counter
length = 0

# Iterate through each character in the string
for char in S:
length += 1

# Print the length
print(length)
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

10

Compilation Status: Passed

Execution Time:

0.014s

TestCase2:

Input:

< hidden >

Expected Output:

GOKUL (gokulmahendran46@gmail.com)

< hidden >

Output:

5

Compilation Status: Passed

Execution Time:

0.009s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.011s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

8

Compilation Status: Passed

Execution Time:

0.014s

TestCase5:

Input:

< hidden >

GOKUL (gokulmahendran46@gmail.com)

Expected Output:

< hidden >

Output:

4

Compilation Status: Passed

Execution Time:

0.013s

37. Problem Statement:Given 2 strings S1 and s2, check whether they are case senitively equal without using any predefined function(case sensitive).If they are not same print 'no'

Sample Input:guvi guvi

Sample Output:yes

Completion Status: Completed

Concepts Included:

strings

array

Language Used: PYTHON 3

GOKUL (gokulmahendran46@gmail.com)

Source Code:

```
# Read input
S1, S2 = input().split()

# First check if lengths are the same
length1 = length2 = 0

for _ in S1:
length1 += 1
for _ in S2:
length2 += 1

if length1 != length2:
print("no")
else:
# Compare character by character
equal = True
for i in range(length1):
```

```
if S1[i] != S2[i]:
equal = False
break
if equal:
print("yes")
else:
print("no")
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.014s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.013s

TestCase3:

Input:

< hidden >

GOKUL (gokulmahendran46@gmail.com)

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.014s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.014s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.013s

GOKUL (gokulmahendran46@gmail.com)

38. Problem Statement:Given a string 'S' and a character 'K', find at

what position the character 'K' occurs for the first time in 'S'. (Assume the index of string starts at 1).If the character is not four in 'S' then print -1

Input Description:Input Size : |s| <= 100000

Sample Input:codekata a

Sample Output:6

Completion Status: Completed

Concepts Included:

strings

Language Used: PYTHON 3

Source Code:

```
# Read input
S, K = input().split()

# Initialize position
position = -1

# Iterate through the string
for i in range(0, 100000):
if i >= len(S):  # Stop if index exceeds string length
break
if S[i] == K:
position = i + 1  # +1 because index starts at 1
break

print(position)
```

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.014s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

GOKUL (gokulmahendran46@gmail.com)

3

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

0.014s

39. Problem Statement:Given a string 'S' and a character 'K', find how many times 'K' got repeated in 'S'.If 'K' is not found in 'S' print -1

Input Description:The input consists of a string 'S' and a character 'K'. The size of string 'S' is at most 100000.

Output Description:The output is the count of character 'K' in string 'S'. If 'K' is not found, print -1.

Sample Input:codekata a

Sample Output:2

Completion Status: Completed

Concepts Included:

strings

Language Used: PYTHON 3

Source Code:

Read input

GOKUL (gokulrajendran46@gmail.com)

```
S, K = input().split()

# Initialize counter
count = 0

# Iterate through each character
for char in S:
if char == K:
count += 1

# Check if character was found
if count == 0:
print(-1)
else:
print(count)
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

2

Compilation Status: Passed

Execution Time:

0.009s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

GOKUL (gokulmahendran46@gmail.com)

0.014s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

3

Compilation Status: Passed

Execution Time:

0.013s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.014s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

5

Compilation Status: Passed

GOKUL (gokulmahendran46@gmail.com)

Execution Time:

0.013s

40. Problem Statement:Given a sentence and string S, find how many times S occurs in the given sentence.If S is not found in the sentence print -1

Input Description:Input Size : |sentence| <= 1000000(complexity O(n)).

Output Description:The output is the number of times S occurs in the given sentence, or -1 if S is not found.

Sample Input:I enjoy doing codekatacodekata

Sample Output:1

Completion Status: Completed

Concepts Included:

strings

array

Language Used: PYTHON 3

Source Code:

```
# Read input
sentence = input()
S = input()

# Initialize counters
count = 0
i = 0
len_sentence = 0
len_S = 0

# Manually calculate lengths (without using len())
for _ in sentence:
len_sentence += 1
for _ in S:
len_S += 1

# Iterate through the sentence
while i <= len_sentence - len_S:
match = True
for j in range(len_S):
if sentence[i + j] != S[j]:
match = False
```

GOKUL (gokulmahendran46@gmail.com)

```
break
if match:
count += 1
i += len_S  # Move past this occurrence
else:
i += 1

# Print result
if count == 0:
print(-1)
else:
print(count)
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

0.013s

GOKUL (gokulmahendran46@gmail.com)

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.014s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

0.012s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

3

Compilation Status: Passed

Execution Time:

GOKUL (gokulmahendran46@gmail.com)

0.011s

41. Problem Statement:Given a number N followed by N elements the number '0' occurs, print the proceeding numbers until the next '0' is encountered. If there are no balancing 0's, print -1.

Input Description:The input consists of an integer N, followed by N elements. N is constrained such that 1 < N <= 100000.

Output Description:Print the numbers between the first two occurrences of '0'. If there are no two '0's, print -1.

Sample Input:101 1 1 0 1 0 1 1 0 1

Sample Output:1 1 1

Completion Status: Not Completed

Concepts Included:

array

Language Used: PYTHON 3

Source Code:

```
def print_between_zeros():
# Read input
N = int(input().strip())
arr = list(map(int, input().split()))

# Find indices of zeros
zero_indices = [i for i, val in enumerate(arr) if val == 0]

# Check if we have at least two zeros
if len(zero_indices) < 2:
print(-1)
else:
# Print elements before the first zero
first_zero = zero_indices[0]
second_zero = zero_indices[1]
result = arr[:first_zero]
if result:
print(*result)
else:
print(-1)
```

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

Accepted

Compilation Status: Failed

Execution Time:

0.014s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

Accepted

Compilation Status: Failed

Execution Time:

0.014s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

Accepted

Compilation Status: Failed

Execution Time:

GOKUL (gokulmahendran46@gmail.com)

0.013s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

Accepted

Compilation Status: Failed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

Accepted

Compilation Status: Failed

Execution Time:

0.014s

GOKUL (gokulmahendran46@gmail.com)

42. Problem Statement:Given 2 strings,check whether they have any common characters.If found print 'yes' else print 'no'.

Input Description:Input Size : |s| <= 100000(O(n))

Sample Input:guvi guvigeeks

Sample Output:yes

Completion Status: Completed

Concepts Included:

strings

Language Used: PYTHON 3

Source Code:

```
# Read input
S1, S2 = input().split()

# Initialize a dictionary to track characters in S1
char_map = {}

# Add characters of S1 to char_map
for char in S1:
char_map[char] = True

# Check if any character in S2 exists in S1
found = False
for char in S2:
if char in char_map:
found = True
break

# Print result
if found:
print("yes")
else:
print("no")
```

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.014s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.014s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

GOKUL (gokulmahendran46@gmail.com)

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.011s

GOKUL (gokumahendran16@gmail.com)

43. Problem Statement:Given 2 numbers N,X and an array of N elements, check if there exists any 2 numbers in the array with sum equal to X.If found print 'yes' otherwise print 'no'

Input Description:The input consists of two numbers N and X, and an array of N elements. N and X are up to 100000.

Output Description:Print 'yes' if two numbers with sum equal to X are found in the array, otherwise print 'no'.

Sample Input:4 42 2 0 0

Sample Output:yes

Completion Status: Completed

Concepts Included:

array

Language Used: PYTHON 3

Source Code:

```
# Read input
N, X = map(int, input().split())
arr = list(map(int, input().split()))

# Initialize a set to store numbers seen so far
seen = set()
```

```
found = False

for num in arr:
if X - num in seen:
found = True
break
seen.add(num)

# Print result
if found:
print("yes")
else:
print("no")
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.011s

GOKUL (gokulmahendran46@gmail.com)

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.014s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.013s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

GOKUL (gokulmahendran46@gmail.com)

0.01s

44. Problem Statement:Given number N, find the minimum factor which yeilds odd number as the quotient.

Input Description:The input consists of a single integer N, where N <= 100000.

Output Description:The output is the minimum factor of N that yields an odd number as the quotient.

Sample Input:9

Sample Output:1

Completion Status: Completed

Concepts Included:

mathematics

Language Used: PYTHON 3

Source Code:

```
# Read input
N = int(input())

factor = 1
temp = N

# Divide by 2 until quotient is odd
while temp % 2 == 0:
factor *= 2
temp //= 2

print(factor)
```

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

4

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

4

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

2

Compilation Status: Passed

Execution Time:

0.014s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

GOKUL (gokulmahendran46@gmail.com)

Output:

1

Compilation Status: Passed

Execution Time:

0.013s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.01s

GOKUL (gokulmahendran46@gmail.com)

45. Problem Statement:Given a number N and 2 arrays A and B of sorted order of size N, print the common elements.If it is not found print -1.

Input Description:Input Size : 1 <= N <= 100000

Sample Input:51 1 1 1 11 2 3 4 5

Sample Output:1

Completion Status: Completed

Concepts Included:

array

Language Used: PYTHON 3

Source Code:

```
# Read input
N = int(input())
```

```
A = list(map(int, input().split()))
B = list(map(int, input().split()))

i = j = 0
common = []

while i < N and j < N:
if A[i] == B[j]:
# Add to common only if not already added
if not common or common[-1] != A[i]:
common.append(A[i])
i += 1
j += 1
elif A[i] < B[j]:
i += 1
else:
j += 1

# Print result
if common:
print(*common)
else:
print(-1)
```

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

0.014s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1 4 8

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1 2

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1 8

Compilation Status: Passed

Execution Time:

0.014s

TestCase5:

Input:

< hidden >

Expected Output:

GOKUL (gokulmahendran46@gmail.com)

< hidden >

Output:

82

Compilation Status: Passed

Execution Time:

0.014s

46. Problem Statement:Given 2 numbers N,K followed by N elements print all the elements lesser than K in sorted order.If the elements could not be found print -1

Input Description:Input Size : N <= 100000

Sample Input:5 3 1 2 1 4 1

Sample Output:1 1 1 2

Completion Status: Not Completed

Concepts Included:

array

Language Used: PYTHON 3

Source Code:

GOKUL (gokulmahendran46@gmail.com)

```
# Read input
inputs = list(map(int, input().split()))
N = inputs[0]
K = inputs[1]
arr = inputs[2:]

# Initialize list for elements less than K
less_than_k = []

# Collect elements manually
for num in arr:
if num < K:
less_than_k.append(num)

# Check if any elements found
if not less_than_k:
print(-1)
else:
# Sort manually using built-in sort
```

```
less_than_k.sort()
for num in less_than_k:
print(num, end=' ')
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Failed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

GOKUL (gokulmahendran46@gmail.com)

Output:

-1

Compilation Status: Failed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Failed

Execution Time:

0.011s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Failed

Execution Time:

0.013s

GOKUL (gokulmahendran46@gmail.com)

47. Problem Statement:Given a number N and an array of N elements, print all elements lesser than N in descending order. If no element found print -1.

Input Description:The input consists of a number N, and an array of N elements. N is

between 1 and 10000 (inclusive).

Output Description:Print all elements from the array that are lesser than N, in descending order. If no such elements are found, print -1.

Sample Input:52 14 15 14 3

Sample Output:3 2

Completion Status: Completed

Concepts Included:

array

Language Used: PYTHON 3

Source Code:

```
# Read input
N = int(input())
arr = list(map(int, input().split()))

# Filter elements less than N
less_than_N = []
for num in arr:
if num < N:
less_than_N.append(num)

# Check if any elements found
if not less_than_N:
print(-1)
else:
# Sort in descending order
less_than_N.sort(reverse=True)
print(*less_than_N)
```

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

2 1

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

GOKUL (gokulmahendran46@gmail.com)

2 1

Compilation Status: Passed

Execution Time:

0.014s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

0

Compilation Status: Passed

Execution Time:

0.01s

GOKUL (gokulrajendran46@gmail.com)

48. Problem Statement:Given 2 numbers N,K and an array of N elements, print the number(s) that has been repeated K times.Print them in ascending order if there are more than one number to be printed.If no element satisfies the pattern then print -1

Input Description:The input consists of two integers N and K, followed by an array of N elements. N and K are up to 100000.

Output Description:Print the numbers that have been repeated K times in ascending order. If no such element exists, print -1.

Sample Input:5 21 2 4 1 2

Sample Output:1 2

Completion Status: Completed

Concepts Included:

array

Language Used: PYTHON 3

Source Code:

```
from collections import Counter

# Read inputs
N, K = map(int, input().split())
arr = list(map(int, input().split()))

# Count frequencies
freq = Counter(arr)

# Find numbers repeated exactly K times
result = [num for num, count in freq.items() if count == K]

# Print result
if result:
print(*sorted(result))
else:
print(-1)
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.016s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

GOKUL (gokulmahendran46@gmail.com)

Execution Time:

0.012s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

3

Compilation Status: Passed

Execution Time:

0.016s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.016s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

GOKUL (gokulmahendran46@gmail.com)

Compilation Status: Passed

Execution Time:

0.017s

49. Problem Statement:Given a number N, find the factorial of N.

Input Description:The input consists of a single integer N, constrained by 1 <= N <= 25.

Output Description:The output is the calculated factorial of N.

Sample Input:5

Sample Output:120

Completion Status: Completed

Concepts Included:

mathematics

Language Used: PYTHON 3

Source Code:

```
# Read input
N = int(input())

# Calculate factorial
fact = 1
for i in range(1, N + 1):
fact *= i

# Print result
print(fact)
```

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

720

Compilation Status: Passed

Execution Time:

0.013s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

2

Compilation Status: Passed

Execution Time:

0.014s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

GOKUL (gokulmahendran46@gmail.com)

Output:

6

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

24

Compilation Status: Passed

Execution Time:

0.01s

GOKUL (gokulmahendran46@gmail.com)

50. Problem Statement:Given a number N and an array of N elements, find the length of the longest repeating sequence of the elements.If no such sequence is found print -1

Input Description:The input consists of an integer N (where N <= 100000) and an array of N elements.

Sample Input:81 2 2 2 3 4 5 6

Sample Output:3

Completion Status: Completed

Concepts Included:

array

Language Used: PYTHON 3

Source Code:

```
# Read input
```

```
N = int(input())
arr = list(map(int, input().split()))

# Initialize variables
max_len = 1
current_len = 1

# Traverse array
for i in range(1, N):
if arr[i] == arr[i-1]:
current_len += 1
max_len = max(max_len, current_len)
else:
current_len = 1

# If no repeating sequence found, print -1
if max_len == 1:
print(-1)
else:
print(max_len)
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

0.011s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

GOKUL (gokulmahendran46@gmail.com)

2

Compilation Status: Passed

Execution Time:

0.013s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

3

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

2

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

GOKUL (gokulmahendran46@gmail.com)

Output:

2

Compilation Status: Passed

Execution Time:

0.014s

51. Problem Statement:Given 2 numbers N,K print the array after deleting the last K elements.

Input Description:N,K <= 100000

Output Description:The array after deleting the last K elements.

Sample Input:5 41 2 3 4 5

Sample Output:1

Completion Status: Completed

Concepts Included:

array

Language Used: PYTHON 3

Source Code:

```
# Read inputs
N, K = map(int, input().split())
arr = list(map(int, input().split()))

# Slice the array to remove last K elements
result = arr[:N-K]

# Print result
if result:
print(*result)
```

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1 2

Compilation Status: Passed

Execution Time:

0.012s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1 1 1 1

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

GOKUL (gokulmahendran46@gmail.com)

Expected Output:

< hidden >

Output:

11 14

Compilation Status: Passed

Execution Time:

0.014s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1 2 3 4 5

Compilation Status: Passed

Execution Time:

0.01s

GOKUL (gokulmahendran46@gmail.com)

52. Problem Statement:Given a string two strings S1 and S2, remove characters from the S1 which are present in the S2.If S1 becomes empty then print -1

Input Description:Input Size : N <= 100000

Sample Input:GUVI GEEK

Sample Output:UVI

Completion Status: Completed

Concepts Included:

strings

Language Used: PYTHON 3

Source Code:

```
def remove_chars(S1, S2):
# Convert S2 into a set for faster lookup
remove_set = set(S2)

# Build result string by keeping only chars not in S2
result = ''.join([ch for ch in S1 if ch not in remove_set])

# If result is empty, return -1
return result if result else "-1"

# Input
S1, S2 = input().split()
print(remove_chars(S1, S2))
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

valid

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

GOKUL (gokulmahendran46@gmail.com)

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

Fh

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

34

Compilation Status: Passed

GOKUL (gokulmahendran46@gmail.com)

Execution Time:

0.01s

53. Problem Statement:Given a number N followed by N elements for every 2 consecutive numbers print the maximum of the 2.

Input Description:The input consists of an integer N, followed by N elements. N is an integer such that N <= 100000, implying an O(n) time complexity solution is expected.

Output Description:The output is a space-separated sequence of the maximums of every two consecutive numbers from the input.

Sample Input:51 1 3 0 5

Sample Output:1 3 3 5

Completion Status: Not Completed

Concepts Included:

array

Language Used: PYTHON 3

Source Code:

```
def consecutive_max(nums):
result = []
for i in range(len(nums) - 1):
result.append(max(nums[i], nums[i+1]))
return result

# Input
N = int(input())
arr = list(map(int, input().split()))

# Compute and print result
```

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

Accepted

Compilation Status: Failed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

Accepted

Compilation Status: Failed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

Accepted

Compilation Status: Failed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

GOKUL (gokulmahendran46@gmail.com)

< hidden >

Output:

Accepted

Compilation Status: Failed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

Accepted

Compilation Status: Failed

Execution Time:

0.014s

Gokul (gokulmahendran46@gmail.com)

54. Problem Statement:Given an value 'M' follwed by array of M elements in which the elements would have been rotated for certain 'N' times from the intial array representation where all elements are arranged in ascending order.Print the 'N' or print -1 if there is no rotation made or cannot be determined.Note: 1<=N<=length of the given array.

Sample Input:515 18 2 3 6 12

Sample Output:2

Completion Status: Completed

Concepts Included:

array

Language Used: PYTHON 3

Source Code:

```
def find_rotation_count(arr):
# Find index of minimum element
min_index = arr.index(min(arr))

# If min element is at index 0, no rotation
if min_index == 0:
return -1
return min_index

# Input
M = int(input())
arr = list(map(int, input().split()))

# Output
print(find_rotation_count(arr))
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

5

Compilation Status: Passed

GOKUL (gokulmahendran46@gmail.com)

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

4

GOKUL (gokulmahendran46@gmail.com)

Compilation Status: Passed

Execution Time:

0.01s

55. Problem Statement:Given a number N,K followed by array of N elements where the difference between any adjacent elements is 1. Find the position of the given number K.If K not found in the array print -1

Input Description:The input consists of two integers N and K, followed by an array of N elements where the difference between any adjacent elements is 1.

Output Description:The output is the position of the given number K. If K is not found in the array, print -1.

Sample Input:5 13 2 1 2 3

Sample Output:3

Completion Status: Completed

Concepts Included:

array

Language Used: PYTHON 3

Source Code:

```
def find_position(arr, K):
for i in range(len(arr)):
if arr[i] == K:
return i + 1   # 1-based index
return -1

# Input
N, K = map(int, input().split())
arr = list(map(int, input().split()))

# Output
print(find_position(arr, K))
```

Compilation Details:

TestCase1:

Input:

GOKUL (gokulmahendran46@gmail.com)

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

5

Compilation Status: Passed

Execution Time:

0.014s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

7

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

GOKUL (gokulmahendran46@gmail.com)

Input:

< hidden >

Expected Output:

< hidden >

Output:

5

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

3

Compilation Status: Passed

Execution Time:

0.01s

GOKUL (gokulmahendran46@gmail.com)

56. Problem Statement:Given an array A of N elements, count the number of distinct pairs (i,j) such that i < j and A[i] < A[j].If no such pairs can be made print -1

Input Description:The input consists of an integer N, representing the number of elements, followed by N space-separated integers representing the elements of array A.

Output Description:The output is a single integer representing the count of distinct pairs (i,j) such that i < j and A[i] < A[j]. If no such pairs can be made, print -1.

Sample Input:51 2 3 4 5

Sample Output:10

Completion Status: Not Completed

Concepts Included:

array

Language Used: PYTHON 3

Source Code:

```
def count_pairs(arr):
n = len(arr)
count = 0
for i in range(n):
for j in range(i+1, n):
if arr[i] < arr[j]:
count += 1
return count if count > 0 else -1

# Input
N = int(input())
arr = list(map(int, input().split()))

# Output
print(count_pairs(arr))
```

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

0.012s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

7

Compilation Status: Failed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

4

Compilation Status: Failed

Execution Time:

0.014s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

3

Compilation Status: Passed

Execution Time:

0.014s

TestCase5:

Input:

< hidden >

GOKUL (gokulmahendran46@gmail.com)

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.01s

57. Problem Statement:Given an array of N elements which follows either even number or odd number series.There may exists at maximum 1 even number in the odd series or 1 odd number in the even series.Find the different number if exists otherwise print '-1'?

Input Description:Input Size : |N| <= 100000

Sample Input:51 3 4 5 7

Sample Output:4

Completion Status: Not Completed

Concepts Included:

array

Language Used: PYTHON 3

GOKUL (gokulmahendran46@gmail.com)

Source Code:

```
def find_different_number(arr):
# Count odd and even numbers
odd_count = sum(1 for x in arr if x % 2 != 0)
even_count = len(arr) - odd_count

# If majority are odd, return the even one
if odd_count > even_count and even_count == 1:
for x in arr:
if x % 2 == 0:
return x

# If majority are even, return the odd one
if even
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

File "script-3.8.1.py", line 13
if even
^
SyntaxError: invalid syntax

Runtime Error (NZEC)

Compilation Status: Failed

Execution Time:

0.009s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

File "script-3.8.1.py", line 13
if even
^
SyntaxError: invalid syntax

Runtime Error (NZEC)

Compilation Status: Failed

Execution Time:

0.01s

TestCase3:

Input:

GOKUL (gokulmahendran46@gmail.com)

< hidden >

Expected Output:

< hidden >

Output:

File "script-3.8.1.py", line 13
if even
^
SyntaxError: invalid syntax

Runtime Error (NZEC)

Compilation Status: Failed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

File "script-3.8.1.py", line 13
if even
^
SyntaxError: invalid syntax

Runtime Error (NZEC)

Compilation Status: Failed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

GOKUL (gokulmahendran46@gmail.com)

< hidden >

Output:

File "script-3.8.1.py", line 13
if even
^
SyntaxError: invalid syntax

Runtime Error (NZEC)

Compilation Status: Failed

Execution Time:

0.01s

58. Problem Statement:Given an array of N elements,find the maximum length of increasing continuous sub-array.If it is not found print '-1'.

Input Description:Input Size : N <= 100000

Output Description:The maximum length of the increasing continuous sub-array, or '-1' if not found.

Sample Input:51 2 3 2 1

Sample Output:3

GOKUL (gokulmahendran6@gmail.com)

Completion Status: Not Completed

Concepts Included:

array

Language Used: PYTHON 3

Source Code:

```
def find_different_number(arr):
# Count odd and even numbers
odd_count = sum(1 for x in arr if x % 2 != 0)
even_count = len(arr) - odd_count

# If majority are odd, return the even one
if odd_count > even_count and even_count == 1:
for x in arr:
if x % 2 == 0:
return x
```

```
# If majority are even, return the odd one
if even_count > odd_count and odd_count == 1:
for x in arr:
if x % 2 != 0:
return x

# Otherwise, no different number
return -1


# Example usage
N = int(input())
arr = list(map(int, input().split()))
print(find_different_number(arr))
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Failed

Execution Time:

0.011s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Failed

Execution Time:

GOKUL (gokulmahendran46@gmail.com)

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Failed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Failed

Execution Time:

0.011s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

GOKUL (gokulmahendran46@gmail.com)

Execution Time:

0.011s

59. Problem Statement:Given 2 array of size N and M.merge them in sorted order and print it.

Input Description:The input consists of two integers N and M representing the sizes of the arrays, followed by N integers for the first array and M integers for the second array. The constraints are |N||M| <= 100000.

Output Description:The output is the merged sorted array.

Sample Input:5 41 2 3 4 51 2 3 4

Sample Output:1 1 2 2 3 3 4 4 5

Completion Status: Completed

Concepts Included:

array

Language Used: PYTHON 3

Source Code:

```
def merge_sorted_arrays(arr1, arr2):
merged = arr1 + arr2
merged.sort()
return merged

# Input
N, M = map(int, input().split())
arr1 = list(map(int, input().split()))
arr2 = list(map(int, input().split()))

# Output
print(*merge_sorted_arrays(arr1, arr2))
```

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1 2 3 4 5 6 7 8 9

Compilation Status: Passed

Execution Time:

0.014s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1 1 2 2 3 3 4 4 5 5

Compilation Status: Passed

Execution Time:

0.014s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1 1 2 2 3 3 4 5

Compilation Status: Passed

Execution Time:

0.013s

TestCase4:

Input:

< hidden >

GOKUL (gokulmahendran46@gmail.com)

Expected Output:

< hidden >

Output:

0 2 3 5 5 5 5 5

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

0 1 1 2 2 3 4 4 5

Compilation Status: Passed

Execution Time:

0.013s

GOKUL (gokulmahendran46@gmail.com)

60. Problem Statement:Given an array, find the maximum difference between any two elements.

Input Description:Input Size : N <= 1000000(complexity O(n) or O(nlogn))

Sample Input:51 2 3 4 5

Sample Output:4

Completion Status: Completed

Concepts Included:

array

Language Used: PYTHON 3

Source Code:

```
def max_difference(arr):
return max(arr) - min(arr)

# Input
N = int(input())
arr = list(map(int, input().split()))

# Output
print(max_difference(arr))
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

4

Compilation Status: Passed

Execution Time:

0.011s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

0

Compilation Status: Passed

Execution Time:

0.014s

TestCase3:

Input:

GOKUL (gokulmahendran46@gmail.com)

< hidden >

Expected Output:

< hidden >

Output:

3

Compilation Status: Passed

Execution Time:

0.013s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

2

Compilation Status: Passed

Execution Time:

0.015s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

0

Compilation Status: Passed

Execution Time:

0.015s

GOKUL (gokulmahendran46@gmail.com)

61. Problem Statement:Given an array, find the absolute minimum difference between any two elements in the array.

Input Description:Input Size : N <= 1000000(complexity O(n) or O(nlogn))

Sample Input:50 2 3 4 5

Sample Output:1

Completion Status: Completed

Concepts Included:

array

Language Used: PYTHON 3

Source Code:

```
def min_abs_difference(arr):
arr.sort()
min_diff = float('inf')
for i in range(1, len(arr)):
min_diff = min(min_diff, arr[i] - arr[i-1])
return min_diff if min_diff != float('inf') else -1

# Input
N = int(input())
arr = list(map(int, input().split()))

# Output
print(min_abs_difference(arr))
```

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

0

Compilation Status: Passed

Execution Time:

0.015s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

0

Compilation Status: Passed

Execution Time:

0.014s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

3

Compilation Status: Passed

GOKUL (gokulmahendran46@gmail.com)

Execution Time:

0.011s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

0

Compilation Status: Passed

Execution Time:

0.01s

62. Problem Statement:Given a number N and an array of N elements, find the maximum of the elements (using Bitwise AND) and print the output.

Input Description:Input Size N <= 100000

Sample Input:42 4 4 2

Sample Output:4

GOKUL (gokulmahendran94@gmail.com)

Completion Status: Completed

Concepts Included:

bitwise

Language Used: PYTHON 3

Source Code:

```
def max_element_bitwise_and(arr):
# The maximum element itself is the answer
return max(arr)

# Input
N = int(input())
arr = list(map(int, input().split()))
```

```
# Output
print(max_element_bitwise_and(arr))
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

4

Compilation Status: Passed

Execution Time:

0.014s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

99

Compilation Status: Passed

Execution Time:

0.013s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

GOKUL (gokulmahendran46@gmail.com)

Output:

2

Compilation Status: Passed

Execution Time:

0.014s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

9

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

345

Compilation Status: Passed

Execution Time:

0.01s

GOKUL (gokulmahendran46@gmail.com)

63. Problem Statement:Given a string S, print the reverse of the string.

Input Description:Input Size : |s| <= 100000 (ie do it in O(n) or O(log n) time complexity)

Sample Input:codekata

Sample Output:atakedoc

Completion Status: Completed

Concepts Included:

strings

Language Used: PYTHON 3

Source Code:

```
def reverse_string(s):
# Python slicing gives O(n) reversal
return s[::-1]

# Input
s = input().strip()

# Output
print(reverse_string(s))
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

i

Compilation Status: Passed

Execution Time:

0.014s

TestCase2:

Input:

< hidden >

GOKUL (gokulmahendran46@gmail.com)

Expected Output:

< hidden >

Output:

skeeg

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

654321

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

cba

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

GOKUL (gokulmahendran46@gmail.com)

< hidden >

Expected Output:

< hidden >

Output:

a

Compilation Status: Passed

Execution Time:

0.01s

64. Problem Statement:Given a String S,print the number of unique characters in it.If all the characters are duplicated,then print -1.

Sample Input:GUVIGEEK

Sample Output:4

Completion Status: Completed

Concepts Included:

mathematics

Language Used: PYTHON 3

Source Code:

GOKUL (gokulmahendran46@gmail.com)

```
from collections import Counter

def count_unique_chars(s):
freq = Counter(s)
unique_count = sum(1 for ch in freq if freq[ch] == 1)
return unique_count if unique_count > 0 else -1

# Input
s = input().strip()

# Output
print(count_unique_chars(s))
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

4

Compilation Status: Passed

Execution Time:

0.011s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

0.016s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

2

Compilation Status: Passed

Execution Time:

0.017s

GOKUL (gokulmahendran46@gmail.com)

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

0.012s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

5

Compilation Status: Passed

Execution Time:

0.012s

GOKUL (gokulmahendran46@gmail.com)

65. Problem Statement:Given a string S consisting of a sentence, the task is to reverse every word of the sentence except the first and last character of the words.

Input Description:The input consists of a string S representing a sentence.

Output Description:The output is the modified string with every word reversed except its first and last characters.

Sample Input:guvi coding platform

Sample Output:gvui cnidog proftalm.

Completion Status: Completed

Concepts Included:

strings

array

Language Used: PYTHON 3

Source Code:

```
def reverse_inner(word):
# If word length <= 2, return as is
if len(word) <= 2:
return word
# Keep first and last, reverse middle
return word[0] + word[1:-1][::-1] + word[-1]

def process_sentence(sentence):
words = sentence.split()
return ' '.join(reverse_inner(word) for word in words)

# Input
s = input().strip()

# Output
print(process_sentence(s))
```

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

i eojny ctakedoa

Compilation Status: Passed

Execution Time:

0.014s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

p c

Compilation Status: Passed

Execution Time:

0.014s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

cdoe

Compilation Status: Passed

Execution Time:

0.014s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

darocem

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

GOKUL (gokulmahendran46@gmail.com)

< hidden >

Expected Output:

< hidden >

Output:

14325

Compilation Status: Passed

Execution Time:

0.011s

66. Problem Statement:Given a roman numeral N, convert it into integer.Take L=50, C=100.If it is not a valid roman numeral print '-1'

Input Description:Input Size : N <= 100

Sample Input:VIY

Sample Output:6-1

Completion Status: Completed

Concepts Included:

strings

Amazon

Facebook

Microsoft

Twitter

Zoho

guvi-learning-path

Language Used: PYTHON 3

Source Code:

```
def roman_to_int(s: str) -> int:
roman_map = {
'I': 1, 'V': 5, 'X': 10,
'L': 50, 'C': 100, 'D': 500, 'M': 1000
}

# Check validity: all characters must be in roman_map
```

GOKUL (gokulmahendran46@gmail.com)

```
for ch in s:
if ch not in roman_map:
return -1

total = 0
prev_value = 0

for ch in reversed(s):
value = roman_map[ch]
if value < prev_value:
total -= value
else:
total += value
prev_value = value

return total


# Driver code
N = input().strip()
print(roman_to_int(N))
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

10

Compilation Status: Passed

Execution Time:

0.013s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

GOKUL (gokulmahendran46@gmail.com)

-1

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

54

Compilation Status: Passed

Execution Time:

0.015s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

95

Compilation Status: Passed

Execution Time:

0.014s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

GOKUL (gokulmahendran46@gmail.com)

Output:

87

Compilation Status: Passed

Execution Time:

0.016s

67. Problem Statement:Given 2 strings,check whether it is isomorphic.If it is not isomorphic print '-1'.

Input Description:The input consists of two strings. The size of each string |s| is at most 100000.

Output Description:The output is 'yes' if the strings are isomorphic, otherwise '-1'.

Sample Input:aab xxy

Sample Output:yes

GOKUL (gokulmahendran46@gmail.com)

Completion Status: Not Completed

Concepts Included:

strings

array

Language Used: PYTHON 3

Source Code:

```
def is_isomorphic(s: str, t: str) -> str:
if len(s) != len(t):
return "-1"

mapping_st = {}
mapping_ts = {}

for c1, c2 in zip(s, t):
if c1 in mapping_st and mapping_st[c1] != c2:
return "-1"
if c2 in mapping_ts and mapping_ts[c2] != c1:
return "-1"
mapping_st[c1] = c2
mapping_ts[c2] = c1

return "yes"
```

```
# Driver code
s, t = input().split()
print(is_isomorphic(s, t))
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Failed

Execution Time:

0.013s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.016s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

GOKUL (gokulmahendran46@gmail.com)

Output:

yes

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Failed

Execution Time:

0.013s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.01s

GOKUL (gokulmahendran46@gmail.com)

68. Problem Statement:Given a string 'S' swap the even and odd characters starting from index 1(Assume the index starts from 0).

Input Description:Input Size : |s| <= 10000000(complexity O(n))

Sample Input:codekata

Sample Output:ocedakat

Completion Status: Completed

Concepts Included:

basics

array

strings

Language Used: PYTHON 3

Source Code:

```
def swap_even_odd(s: str) -> str:
chars = list(s)  # convert to list for easy swapping
n = len(chars)

for i in range(0, n - 1, 2):  # step by 2
chars[i], chars[i+1] = chars[i+1], chars[i]

return "".join(chars)


# Driver code
S = input().strip()
print(swap_even_odd(S))
```

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

ugiv

Compilation Status: Passed

Execution Time:

0.015s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

ejardl

Compilation Status: Passed

Execution Time:

0.011s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

2143

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

egkes

Compilation Status: Passed

Execution Time:

GOKUL (gokulmahendran46@gmail.com)

0.013s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

oced

Compilation Status: Passed

Execution Time:

0.014s

69. Problem Statement:Given a string/sentence print its corresponding camelcase convention.

Input Description:Input Size : |s| <= 1000000(complexity O(n))

Sample Input:guvi geeks

Sample Output:GuviGeeks

Completion Status: Completed

Concepts Included:

strings

Language Used: PYTHON 3

Source Code:

```
def to_camel_case(s: str) -> str:
words = s.split()
# Capitalize each word and join
return "".join(word.capitalize() for word in words)

# Driver code
S = input().strip()
print(to_camel_case(S))
```

Compilation Details:

GOKUL (gokulmahendran46@gmail.com)

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

G

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

Codekata

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

GuviIsForCoders

Compilation Status: Passed

Execution Time:

GOKUL (gokulmahendran46@gmail.com)

0.014s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

MicroArc

Compilation Status: Passed

Execution Time:

0.014s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

AlphaBeta

Compilation Status: Passed

Execution Time:

0.01s

GOKUL (gokulmahendran46@gmail.com)

70. Problem Statement:Given a range of 2 numbers (i.e) L and R count the number of prime numbers in the range (inclusive of L and R).

Input Description:Input Size : L <= R <= 100000(complexity O(n) read about Sieve of Eratosthenes)

Sample Input:2 5

Sample Output:3

Completion Status: Not Completed

Concepts Included:

mathematics

basics

Language Used: PYTHON 3

Source Code:

Problem Statement:
Given a range of 2 numbers (i.e) L and R count the number of prime numbers in the range (inclusive of L and R).

Input Description:
Input Size : L <= R <= 100000(complexity O(n) read about Sieve of Eratosthenes)

Sample Input:
2 5

Sample Output:
3

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

File "script-3.8.1.py", line 1
Problem Statement:
^
SyntaxError: invalid syntax

Runtime Error (NZEC)

Compilation Status: Failed

Execution Time:

0.009s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

File "script-3.8.1.py", line 1
Problem Statement:
^
SyntaxError: invalid syntax

Runtime Error (NZEC)

Compilation Status: Failed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

File "script-3.8.1.py", line 1
Problem Statement:
^
SyntaxError: invalid syntax

Runtime Error (NZEC)

Compilation Status: Failed

Execution Time:

0.01s

TestCase4:

Input:

GOKUL (gokulmahendran46@gmail.com)

< hidden >

Expected Output:

< hidden >

Output:

File "script-3.8.1.py", line 1
Problem Statement:
^
SyntaxError: invalid syntax

Runtime Error (NZEC)

Compilation Status: Failed

Execution Time:

0.014s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

File "script-3.8.1.py", line 1
Problem Statement:
^
SyntaxError: invalid syntax

Runtime Error (NZEC)

Compilation Status: Failed

Execution Time:

0.009s

GOKUL (gokulmahendran46@gmail.com)

71. Problem Statement:Given 2 strings check whether they differ exacly by one character.If yes then print 'yes' otherwise print 'no'

Input Description:Input Size : |s| <= 100000(complexity O(nlogn) or O(n))

Sample Input:codekata codekate

Sample Output:yes

Completion Status: Completed

Concepts Included:

strings

loop

Language Used: PYTHON 3

Source Code:

```
def differ_by_one(s1: str, s2: str) -> str:
if len(s1) != len(s2):
return "no"

diff_count = 0
for c1, c2 in zip(s1, s2):
if c1 != c2:
diff_count += 1
if diff_count > 1:
return "no"

return "yes" if diff_count == 1 else "no"


# Driver code
s1, s2 = input().split()
print(differ_by_one(s1, s2))
```

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.014s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.014s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.014s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

GOKUL (gokulmahendran46@gmail.com)

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.014s

72. Problem Statement:Given a day, print 'yes' if it is a holiday otherwise print'no'.Assume that weekend days are holidays

Sample Input:saturdaymonday

Sample Output:yesno

Completion Status: Completed

Concepts Included:

strings

Language Used: PYTHON 3

Source Code:

```
def is_holiday(day: str) -> str:
# Normalize to lowercase for comparison
day = day.lower()
if day in ["saturday", "sunday"]:
return "yes"
else:
return "no"


# Driver code
```

GOKUL (gokulmahendra14@gmail.com)

```
day = input().strip()
print(is_holiday(day))
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.014s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

GOKUL (gokulmahendran46@gmail.com)

no

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.014s

GOKUL (gokulrajendran46@gmail.com)

73. Problem Statement:Given two numbers N,K followed by an array of N elements, print the array after doing right shift K times (in cyclic manner).

Input Description:The input consists of two numbers N, K, followed by an array of N elements. Input Size: 1 <= N, K <= 100000.

Output Description:The output is the array after performing a right cyclic shift K times.

Sample Input:3 27 2 3

Sample Output:2 3 7

Completion Status: Completed

Concepts Included:

mathematics

array

Language Used: PYTHON 3

Source Code:

```
def right_shift_array(arr, N, K):
# Effective shifts
K = K % N
# Perform rotation using slicing
return arr[-K:] + arr[:-K]

# Driver code
N, K = map(int, input().split())
arr = list(map(int, input().split()))

result = right_shift_array(arr, N, K)
print(*result)
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

2 1

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

2 1

Compilation Status: Passed

Execution Time:

0.011s

GOKUL (gokulmahendran46@gmail.com)

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

4 5 1 2 3

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

2 4

Compilation Status: Passed

Execution Time:

0.01s

GOKUL (gokulmahendran46@gmail.com)

74. Problem Statement:Given a number N, print the sum of the squares of its digits.

Input Description:The input consists of a number N, where 1 <= N <= 10000000000000000000.

Output Description:The output is the sum of the squares of the digits of N.

Sample Input:19

Sample Output:82

Completion Status: Completed

Concepts Included:

mathematics

strings

Language Used: PYTHON 3

Source Code:

```
def sum_of_squares(N: str) -> int:
total = 0
for digit in N:
total += int(digit) ** 2
return total

# Driver code
N = input().strip()
print(sum_of_squares(N))
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

90

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

GOKUL (gokulmahendran46@gmail.com)

20

Compilation Status: Passed

Execution Time:

0.013s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

7

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

2

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

GOKUL (gokulmahendran46@gmail.com)

Output:

1

Compilation Status: Passed

Execution Time:

0.014s

75. Problem Statement:Given a string S, print the reverse of the string after removing the vowels.If the resulting string is empty print '-1'.

Input Description:Input Size : 1 <= N <= 100000

Sample Input:codekata

Sample Output:tkdc

Completion Status: Completed

Concepts Included:

strings

array

Language Used: PYTHON 3

Source Code:

```
def reverse_without_vowels(s: str) -> str:
vowels = set("aeiouAEIOU")
# Remove vowels
filtered = [ch for ch in s if ch not in vowels]
# Reverse the filtered string
result = "".join(filtered[::-1])
return result if result else "-1"

# Driver code
S = input().strip()
print(reverse_without_vowels(S))
```

Compilation Details:

TestCase1:

Input:

GOKUL (gokulmahendran46@gmail.com)

< hidden >

Expected Output:

< hidden >

Output:

mhtyhr

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

dctnrl

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

GOKUL (gokulmahendran46@gmail.com)

Input:

< hidden >

Expected Output:

< hidden >

Output:

%$!@4#

Compilation Status: Passed

Execution Time:

0.014s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

10

Compilation Status: Passed

Execution Time:

0.009s

GOKUL (gokulmahendran46@gmail.com)

76. Problem Statement:Given a string S,count the maximum number of times a character repeated in the string.If no character is repeated print '0'.

Input Description:Input Size : 1 <= N <= 100000

Sample Input:codekata

Sample Output:2

Completion Status: Completed

Concepts Included:

array

strings

Language Used: PYTHON 3

Source Code:

```
from collections import Counter

def max_repeated_char_count(s: str) -> int:
freq = Counter(s)
max_count = max(freq.values())
return max_count if max_count > 1 else 0

# Driver code
S = input().strip()
print(max_repeated_char_count(S))
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

3

Compilation Status: Passed

Execution Time:

0.012s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

2

Compilation Status: Passed

GOKUL (gokulmahendran46@gmail.com)

Execution Time:

0.012s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

4

Compilation Status: Passed

Execution Time:

0.017s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

3

Compilation Status: Passed

Execution Time:

0.016s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

0

GOKUL (gokulmahendran46@gmail.com)

Compilation Status: Passed

Execution Time:

0.012s

77. Problem Statement:Given a number N and an array of N elements, every number is repeated except for one. Print that one number.

Input Description:Input Size : 1 <= N <= 100000

Sample Input:101 2 3 2 3 3 2 5 5 2

Sample Output:1

Completion Status: Not Completed

Concepts Included:

mathematics

array

algorithm

Language Used: PYTHON 3

Source Code:

```
def find_unique(arr):
result = 0
for num in arr:
result ^= num
return result

# Driver code
N = int(input().strip())
arr = list(map(int, input().split()))
print(find_unique(arr))
```

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.014s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

3

Compilation Status: Failed

Execution Time:

0.011s

TestCase4:

Input:

< hidden >

GOKUL (gokulmahendran46@gmail.com)

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.013s

78. Problem Statement:Given two numbers L,R print the smallest number which is divisible by both L and R.

Input Description:The input consists of two numbers L and R, where 1 <= L,R <= 100000.

Output Description:The output is the smallest number which is divisible by both L and R.

Sample Input:10 130

Sample Output:130

GOKUL (gokulmahendran46@gmail.com)

Completion Status: Completed

Concepts Included:

mathematics

array

Language Used: PYTHON 3

Source Code:

```
import math

def smallest_divisible(L, R):
return (L * R) // math.gcd(L, R)

# Driver code
L, R = map(int, input().split())
print(smallest_divisible(L, R))
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

60

Compilation Status: Passed

Execution Time:

0.011s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

810

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

GOKUL (gokulmahendran46@gmail.com)

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

792

Compilation Status: Passed

Execution Time:

0.01s

GOKUL (gokulmanohar1@gmail.com)

79. Problem Statement:Given a number N and an array of N strings, find the number of strings that are an anagram of 'kabali'.If there exists no anagram for the given string print '0'.

Input Description:The input consists of an integer N, representing the number of strings, followed by N strings.Constraints: 1 <= N <= 1000

Output Description:The output is a single integer representing the count of strings that are an anagram of 'kabali', or '0' if no such anagram exists.

Sample Input:5kabalikaablikababakabkabail

Sample Output:3

Completion Status: Completed

Concepts Included:

array

strings

Language Used: PYTHON 3

Source Code:

```
from collections import Counter

def count_anagrams(strings, target="kabali"):
target_count = Counter(target)
```

```
count = 0
for s in strings:
if len(s) == len(target) and Counter(s) == target_count:
count += 1
return count if count > 0 else 0

# Driver code
N = int(input().strip())
arr = [input().strip() for _ in range(N)]
print(count_anagrams(arr))
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

3

Compilation Status: Passed

Execution Time:

0.012s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

0

Compilation Status: Passed

Execution Time:

0.012s

TestCase3:

GOKUL (gokulmahendran46@gmail.com)

Input:

< hidden >

Expected Output:

< hidden >

Output:

3

Compilation Status: Passed

Execution Time:

0.017s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.012s

GOKUL (gokulmahendran46@gmail.com)

80. Problem Statement:Given a number N, print their prime factors in sorted order.

Input Description:The input consists of a number N, where 2 <= N <= 100000.

Sample Input:18

Sample Output:2 3

Completion Status: Completed

Concepts Included:

algorithm

array

mathematics

Language Used: PYTHON 3

Source Code:

```
def prime_factors(n: int):
factors = []

# Check for factor 2
if n % 2 == 0:
factors.append(2)
while n % 2 == 0:
n //= 2

# Check for odd factors
i = 3
while i * i <= n:
if n % i == 0:
factors.append(i)
while n % i == 0:
n //= i
i += 2

# If n is still > 2, then it's prime
if n > 2:
factors.append(n)

return sorted(factors)


# Driver code
N = int(input().strip())
result = prime_factors(N)
print(*result)
```

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

5

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

2 3

Compilation Status: Passed

Execution Time:

0.014s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

2 5

Compilation Status: Passed

Execution Time:

0.014s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

19

GOKUL (gokulmahendran46@gmail.com)

Compilation Status: Passed

Execution Time:

0.015s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

3 5 7

Compilation Status: Passed

Execution Time:

0.01s

GOKUL (gokulmanoharan46@gmail.com)

81. Problem Statement:Given a string S, print the encoded string by adding 3 to each character(a maps to d,b maps to e,c maps to f and so on).

Input Description:Input Size : 1 <= N <= 100000

Sample Input:RADAR

Sample Output:UDGDU

Completion Status: Completed

Concepts Included:

strings

array

Language Used: PYTHON 3

Source Code:

```
def encode_string(s: str) -> str:
result = []
for ch in s:
if 'a' <= ch <= 'z':  # lowercase
```

```
result.append(chr((ord(ch) - ord('a') + 3) % 26 + ord('a')))
elif 'A' <= ch <= 'Z':  # uppercase
result.append(chr((ord(ch) - ord('A') + 3) % 26 + ord('A')))
else:
# Non-alphabet characters remain unchanged
result.append(ch)
return "".join(result)


# Driver code
S = input().strip()
print(encode_string(S))
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

ARUR

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

MLPPLH

Compilation Status: Passed

Execution Time:

0.013s

GOKUL (gokulmahendran46@gmail.com)

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

MRN

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

CCCCC

Compilation Status: Passed

Execution Time:

0.01s

GOKUL (gokulmahendran46@gmail.com)

82. Problem Statement:Given 3 points check whether they lie on the same line.If they lie on the same line print 'yes' Otherwise print 'no'.

Sample Input:0 10 00 2

Sample Output:yes

Completion Status: Completed

Concepts Included:

mathematics

array

algorithm

Language Used: PYTHON 3

Source Code:

```
def are_collinear(x1, y1, x2, y2, x3, y3):
area = x1 * (y2 - y3) + x2 * (y3 - y1) + x3 * (y1 - y2)
return "yes" if area == 0 else "no"

# Driver code
x1, y1 = map(int, input().split())
x2, y2 = map(int, input().split())
x3, y3 = map(int, input().split())

print(are_collinear(x1, y1, x2, y2, x3, y3))
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.012s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

GOKUL (gokulmahendran46@gmail.com)

Execution Time:

0.013s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.012s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.014s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

GOKUL (gokulmahendran46@gmail.com)

Compilation Status: Passed

Execution Time:

0.01s

83. Problem Statement:Given a number N and a number K, find the greatest number which divides both.

Input Description:N and K <= 100000

Sample Input:5 10

Sample Output:5

Completion Status: Completed

Concepts Included:

mathematics

Language Used: PYTHON 3

Source Code:

```
import math

def greatest_divisor(N, K):
return math.gcd(N, K)

# Driver code
N, K = map(int, input().split())
print(greatest_divisor(N, K))
```

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

5

Compilation Status: Passed

Execution Time:

0.014s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

2

Compilation Status: Passed

Execution Time:

0.011s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

10

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

GOKUL (gokulmahendran46@gmail.com)

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

12

Compilation Status: Passed

Execution Time:

0.014s

84. Problem Statement:Given a number N and K followed by N elements and K elements. Now insert the given K elements one by one into the array and print the maximum in the array after each insertion .

Input Description:Input Size : K <= N <= 10000(read about priority queues and implement)

Sample Input:5 21 2 3 4 55 4

Sample Output:5 5

GOKUL (gokulkarthik00n46@gmail.com)

Completion Status: Completed

Concepts Included:

array

data structures

algorithm

Language Used: PYTHON 3

Source Code:

```
def insert_and_get_max(arr, inserts):
current_max = max(arr)   # initial maximum
result = []
for x in inserts:
arr.append(x)        # insert element
if x > current_max:  # update max if needed
current_max = x
result.append(current_max)
return result


# Driver code
N, K = map(int, input().split())
arr =list(map(int, input().split()))
inserts = list(map(int, input().split()))

output = insert_and_get_max(arr, inserts)
print(*output)
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

5 5

Compilation Status: Passed

Execution Time:

0.013s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

5 5

GOKUL (gokulmahendran46@gmail.com)

Compilation Status: Passed

Execution Time:

0.014s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

10 20 20

Compilation Status: Passed

Execution Time:

0.014s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

5 6

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

GOKUL (gokulmahendran46@gmail.com)

5 10

Compilation Status: Passed

Execution Time:

0.01s

85. Problem Statement:Given a string S.Validate if a given string is numeric.print 'yes' if it is a numeric otherwise print 'no'.

Sample Input:guvigeeks

Sample Output:no

Completion Status: Completed

Concepts Included:

mathematics

strings

Language Used: PYTHON 3

Source Code:

```
def is_numeric(s: str) -> str:
return "yes" if s.isdigit() else "no"

# Driver code
S = input().strip()
print(is_numeric(S))
```

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.013s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.012s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

GOKUL (gokulmahendran46@gmail.com)

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.014s

86. Problem Statement:Given an array of N strings sort it in ascending order based on the length of the string.If two strings are found to have the same length sort them in lexicographical order.

Sample Input:3coding platform codekata

Sample Output:coding codekata platform

GOKUL (gokulraj.cseen46@gmail.com)

Completion Status: Completed

Concepts Included:

algorithm

Language Used: PYTHON 3

Source Code:

```
def sort_strings(arr):
# Sort by length first, then lexicographically
arr.sort(key=lambda x: (len(x), x))
return arr

# Driver code
N = int(input().strip())
```

```
arr = input().split()
result = sort_strings(arr)
print(*result)
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

aa ab aac

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

aa aac aba

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

GOKUL (gokulmahendran46@gmail.com)

Output:

ab aac aaaa

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

aa

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

a b c

Compilation Status: Passed

Execution Time:

0.01s

GOKUL (gokulmahendran46@gmail.com)

87. Problem Statement:Given a sentence S take out the extra spaces.If no extra space is present print the same as output.

Input Description:Input Size : |s| <= 100000(complexity O(n))

Sample Input:codekata challenge

Sample Output:codekata challenge

Completion Status: Completed

Concepts Included:

array

strings

Language Used: PYTHON 3

Source Code:

```
def remove_extra_spaces(s: str) -> str:
# Split by whitespace and rejoin with single space
return " ".join(s.split())

# Driver code
S = input().rstrip()
print(remove_extra_spaces(S))
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

coding platform

Compilation Status: Passed

Execution Time:

0.014s

TestCase2:

Input:

< hidden >

Expected Output:

GOKUL (gokulmahendran46@gmail.com)

< hidden >

Output:

guvi geeks

Compilation Status: Passed

Execution Time:

0.009s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

aaa BBB

Compilation Status: Passed

Execution Time:

0.015s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

a a v

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

GOKUL (gokulmahendran46@gmail.com)

Expected Output:

< hidden >

Output:

aa bb cc

Compilation Status: Passed

Execution Time:

0.015s

88. Problem Statement:Given a string S change upper case to lowercase and lowercase to uppercase.

Input Description:The input consists of a string S with size |s| <= 10000000 (complexity O(n)).

Sample Input:CodEkaTa

Sample Output:cODeKAtA

GOKUL (gokulmahendran46@gmail.com)

Completion Status: Completed

Concepts Included:

strings

Language Used: PYTHON 3

Source Code:

```
def swap_case(s: str) -> str:
return s.swapcase()

# Driver code
S = input().strip()
print(swap_case(S))
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

ABCd

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

AAAA

Compilation Status: Passed

Execution Time:

0.014s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

BABA

Compilation Status: Passed

Execution Time:

0.013s

TestCase4:

Input:

< hidden >

GOKUL (gokulmahendran46@gmail.com)

Expected Output:

< hidden >

Output:

aABc

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

aabb

Compilation Status: Passed

Execution Time:

0.014s

GOKUL (gokulmahendran46@gmail.com)

89. Problem Statement:Given a string/sentence remove all the spaces and print the result.

Input Description:Input Size : |s| <= 1000000(complexity O(n))

Sample Input:guvi geeks

Sample Output:guvigeeks

Completion Status: Completed

Concepts Included:

strings

Language Used: PYTHON 3

Source Code:

```
def remove_spaces(s: str) -> str:
return s.replace(" ", "")

# Driver code
S = input().rstrip()
print(remove_spaces(S))
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

AabXxy

Compilation Status: Passed

Execution Time:

0.014s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

AbCd

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

GOKUL (gokulmahendran46@gmail.com)

Expected Output:

< hidden >

Output:

AaaBbb

Compilation Status: Passed

Execution Time:

0.014s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

aabc

Compilation Status: Passed

Execution Time:

0.009s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

aabb

Compilation Status: Passed

Execution Time:

0.014s

GOKUL (gokulmahendran46@gmail.com)

90. Problem Statement:Given a range (i.e) two numbers L and R

count the number of perfect squares within the range (inclusive of and R).If no perfect square exists within the range print '-1'.

Input Description:The input consists of two integers L and R, representing the range where L <= R <= 100000.

Output Description:The output is an integer representing the count of perfect squares within the range [L, R], or -1 if none exist.

Sample Input:2 10

Sample Output:2

Completion Status: Completed

Concepts Included:

mathematics

array

Language Used: PYTHON 3

Source Code:

```
import math

def count_perfect_squares(L, R):
start = math.ceil(math.sqrt(L))
end = math.floor(math.sqrt(R))
count = end - start + 1
return count if count > 0 else -1

# Driver code
L, R = map(int, input().split())
print(count_perfect_squares(L, R))
```

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

2

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.011s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

GOKUL (gokulmahendran46@gmail.com)

2

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.01s

91. Problem Statement:Given 2 strings and a number K, check whether they differ exactly by K characters.

Input Description:Input Size : |s| <= 100000(complexity O(nlogn) or O(n))

Sample Input:codekata codeguvi 4

Sample Output:yes

GOKUL (gokulrajendran46@gmail.com)

Completion Status: Not Completed

Concepts Included:

array

strings

Language Used: PYTHON 3

Source Code:

```
import math

def count_perfect_squares(L, R):
```

```
start = math.ceil(math.sqrt(L))
end = math.floor(math.sqrt(R))
count = end - start + 1
return count if count > 0 else -1

# Driver code
L, R = map(int, input().split())
print(count_perfect_squares(L, R))
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

```
Traceback (most recent call last):
File "script-3.8.1.py", line 10, in <module>
L, R = map(int, input().split())
ValueError: invalid literal for int() with base 10: 'guvi'
```

Runtime Error (NZEC)

Compilation Status: Failed

Execution Time:

0.01s

GOKUL (gokulmaheswaran46@gmail.com)

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

```
Traceback (most recent call last):
File "script-3.8.1.py", line 10, in <module>
L, R = map(int, input().split())
ValueError: invalid literal for int() with base 10: 'abc'
```

Runtime Error (NZEC)

Compilation Status: Failed

Execution Time:

0.015s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

Traceback (most recent call last):
File "script-3.8.1.py", line 10, in <module>
L, R = map(int, input().split())
ValueError: invalid literal for int() with base 10: 'aaa'

Runtime Error (NZEC)

Compilation Status: Failed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

Traceback (most recent call last):
File "script-3.8.1.py", line 10, in <module>
L, R = map(int, input().split())
ValueError: invalid literal for int() with base 10: 'aa'

Runtime Error (NZEC)

Compilation Status: Failed

GOKUL (gokulmahendran46@gmail.com)

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

Traceback (most recent call last):
File "script-3.8.1.py", line 10, in <module>
L, R = map(int, input().split())
ValueError: invalid literal for int() with base 10: 'aa'

Runtime Error (NZEC)

Compilation Status: Failed

Execution Time:

0.015s

GOKUL (abulmahendran46@gmail.com)

92. Problem Statement:Given a string S consisting of only '(' and ')', print 'yes' if it is balanced otherwise print 'no'.

Sample Input:(())

Sample Output:yes

Completion Status: Completed

Concepts Included:

strings

array

data structures

Language Used: PYTHON 3

Source Code:

def is_balanced(s: str) -> str:

```
count = 0
for ch in s:
if ch == '(':
count += 1
elif ch == ')':
count -= 1
if count < 0:  # more ')' than '('
return "no"
return "yes" if count == 0 else "no"

# Driver code
S = input().strip()
print(is_balanced(S))
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.013s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.012s

GOKUL (gokulmahendran46@gmail.com)

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.014s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.01s

GOKUL (gokulmahendran46@gmail.com)

93. Problem Statement:Given 2 numbers N,K followed by a sorted array of N elements, search and tell if an element K is present in the array.print 'yes' if element is present otherwise print 'no'.

Input Description:Input Size : 1 <= N <= 1000000000000000(Do it in logN time complexity)

Output Description:print 'yes' if element is present otherwise print 'no'.

Sample Input:3 22 3 7

Sample Output:Yes

Completion Status: Not Completed

Concepts Included:

algorithm

array

Language Used: PYTHON 3

Source Code:

```
def binary_search(arr, k):
low, high = 0, len(arr) - 1
while low <= high:
mid = (low + high) // 2
if arr[mid] == k:
return True
elif arr[mid] < k:
low = mid + 1
else:
high = mid - 1
return False

# Driver code
N, K = map(int, input().split())
arr = list(map(int, input().split()))

if binary_search(arr, K):
print("Yes")
else:
print("No")
```

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

No

Compilation Status: Failed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

Yes

Compilation Status: Failed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

Yes

Compilation Status: Failed

Execution Time:

0.014s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

No

Compilation Status: Failed

Execution Time:

GOKUL (gokulmahendran46@gmail.com)

0.014s

94. Problem Statement:Given a number N, and a N*N matrix containing only 0's and 1's, where 0's represent water and 1's representing land, find the number of island 1's(1 surrounded by 0 on all of its adjacent sides).If the island could not be found with the given matrix print -1.

Input Description:The input consists of a number N, and an N*N matrix containing only 0's and 1's. 0's represent water and 1's represent land. N is between 1 and 1000 (inclusive).

Output Description:The output should be the number of island 1's (1 surrounded by 0 on all of its adjacent sides). If no island could be found, print -1.

Sample Input:31 1 00 0 00 1 0

Sample Output:1

Completion Status: Completed

Concepts Included:

mathematics

Amazon

Citrix

D-E-Shaw

Informatica

Intuit

Linkedin

Microsoft

Ola-Cabs

One97

Opera

OYO-Rooms

Paytm

Samsung

Snapdeal

Streamoid-Technologies

GOKUL (gokulmahendran46@gmail.com)

Visa

guvi-learning-path

Language Used: PYTHON 3

Source Code:

```
def count_islands(matrix, n):
count = 0
for i in range(n):
for j in range(n):
if matrix[i][j] == 1:
# Check all four directions
up    = (i == 0 or matrix[i-1][j] == 0)
down  = (i == n-1 or matrix[i+1][j] == 0)
left  = (j == 0 or matrix[i][j-1] == 0)
right = (j == n-1 or matrix[i][j+1] == 0)

if up and down and left and right:
count += 1
return count if count > 0 else -1


# Driver code
N = int(input().strip())
matrix = [list(map(int, input().split())) for _ in range(N)]

print(count_islands(matrix, N))
```

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

2

Compilation Status: Passed

Execution Time:

0.014s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

5

Compilation Status: Passed

Execution Time:

0.014s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

0.015s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.01s

GOKUL (gokulmahendran46@gmail.com)

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

0.01s

95. Problem Statement:Given 2 numbers N and K.Print the number of occurrences of K in N.If K is not found print '-1'.

Input Description:1 <= N <= 100000, 0 <= K <= 9

Sample Input:1000 0

Sample Output:3

Completion Status: Completed

Concepts Included:

array

Language Used: PYTHON 3

Source Code:

```
def count_occurrences(N, K):
N_str = str(N)
K_str = str(K)
count = N_str.count(K_str)
return count if count > 0 else -1

# Driver code
N, K = map(int, input().split())
print(count_occurrences(N, K))
```

Compilation Details:

GOKUL (gokulmahendran46@gmail.com)

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

0.015s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.015s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

2

Compilation Status: Passed

Execution Time:

GOKUL (gokulmahendran46@gmail.com)

0.013s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

3

Compilation Status: Passed

Execution Time:

0.014s

GOKUL (gokulmahendran46@gmail.com)

96. Problem Statement:Given a string S, print the 1st and 3rd character of the string (chracter index starts from 1).

Input Description:Input Size : 1 <= N <= 100000

Sample Input:codekata

Sample Output:cd

Completion Status: Completed

Concepts Included:

strings

Language Used: PYTHON 3

Source Code:

```
def first_and_third(s: str) -> str:
# Ensure string has at least 3 characters
if len(s) >= 3:
return s[0] + s[2]
elif len(s) >= 1:
return s[0]   # only first character if length < 3
else:
return ""      # empty string case

# Driver code
S = input().strip()
```

print(first_and_third(S))

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

gv

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

vr

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

GOKUL (gokulmahendran46@gmail.com)

02

Compilation Status: Passed

Execution Time:

0.011s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

gv

Compilation Status: Passed

Execution Time:

0.01s

GOKUL (gokulrajendran46@gmail.com)

97. Problem Statement:Given a string, print the least repeated characters in the string.If there are more than one character repeated preserve the order as in the input.

Input Description:Input Size : 1 <= N <= 100000

Sample Input:codeKata challenge

Sample Output:odKthng

Completion Status: Not Completed

Concepts Included:

strings

Language Used: PYTHON 3

Source Code:

```
from collections import Counter

def least_repeated_chars(s: str) -> str:
freq = Counter(s)            # count frequency of each character
```

```
min_freq = min(freq.values())  # find minimum frequency
result = []
for ch in s:
if freq[ch] == min_freq:
result.append(ch)
return "".join(result)

# Driver code
S = input().rstrip()
print(least_repeated_chars(S))
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

He Wd

Compilation Status: Failed

Execution Time:

0.017s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

gguuvvii

Compilation Status: Failed

Execution Time:

0.012s

TestCase3:

GOKUL (gokulmahendran46@gmail.com)

Input:

< hidden >

Expected Output:

< hidden >

Output:

71

Compilation Status: Failed

Execution Time:

0.017s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

9 3

Compilation Status: Failed

Execution Time:

0.012s

TestCase5:

Input:

< hidden >

Expected Output:

< hidden >

Output:

w bay

Compilation Status: Failed

Execution Time:

0.011s

GOKUL (gokulmahendran46@gmail.com)

98. Problem Statement:Vishal is learning to skate on ice. He's a beginner, so his only mode of transportation is pushing off from a snow drift to the north, east, south or west and sliding until he lan in another snow drift. He has noticed that in this way it's impossib to get from some snow drifts to some other by any sequence of moves. He now wants to heap up some additional snow drifts, so that he can get from any snow drift to any other one. He asked you to find the minimal number of snow drifts that need to be created. We assume that Vishal can only heap up snow drifts at integer coordinates.

Input Description:The first line of input contains a single integer N the number of snow drifts. Each of the following n lines contains two integers xi and yi the coordinates of the i-th snow drift. Note that the north direction coincides with the direction of Oy axis, so the east direction coincides with the direction of the Ox axis. All snow drift's locations are distinct. Input Size : 1 <= N,Xi,Yi <= 1000

Output Description:The minimal number of snow drifts that need to be created.

Sample Input:22 11 2

Sample Output:1

GOKUL (gokulmahendran46@gmail.com)

Completion Status: Completed

Concepts Included:

trees and graphs

mathematics

array

Language Used: PYTHON 3

Source Code:

```
from collections import defaultdict, deque

def minimal_new_drifts(n, points):
# Build adjacency list based on same x or y
adj = defaultdict(list)
for i in range(n):
for j in range(i+1, n):
if points[i][0] == points[j][0] or points[i][1] == points[j][1]:
adj[i].append(j)
adj[j].append(i)

visited = [False] * n
components = 0
```

```
# BFS to count connected components
for i in range(n):
if not visited[i]:
components += 1
queue = deque([i])
visited[i] = True
while queue:
node = queue.popleft()
for nei in adj[node]:
if not visited[nei]:
visited[nei] = True
queue.append(nei)

return components - 1


# Driver code
N = int(input().strip())
points = [tuple(map(int, input().split())) for _ in range(N)]
print(minimal_new_drifts(N, points))
```

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

1

Compilation Status: Passed

Execution Time:

0.02s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

0

Compilation Status: Passed

Execution Time:

0.016s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

7

Compilation Status: Passed

Execution Time:

0.021s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

5

Compilation Status: Passed

Execution Time:

0.016s

GOKUL (gokulmahendran46@gmail.com)

99. Problem Statement:Given a number N, print the even factors of N.If the even factor does not exists for N print '-1'.

Input Description:Input Size : 1 <= N <= 1000

Sample Input:8

Sample Output:2 4 8

Completion Status: Completed

Concepts Included:

mathematics

array

Language Used: PYTHON 3

Source Code:

```
def even_factors(n):
result = []
for i in range(2, n+1, 2):  # check only even numbers
if n % i == 0:
result.append(i)
if result:
print(*result)
else:
print(-1)

# Driver code
N = int(input().strip())
even_factors(N)
```

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

0.014s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

2

Compilation Status: Passed

Execution Time:

0.014s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

2 4 8 10 20 40

Compilation Status: Passed

Execution Time:

0.014s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

2 10

Compilation Status: Passed

Execution Time:

0.014s

TestCase5:

GOKUL (gokulmahendran46@gmail.com)

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

0.013s

100. Problem Statement:Given a number N, print the distinct pairs formed by multiplying two prime numbers (i.e)prime x prime should yield the N.Also print the numbers in descending order.If no such pairs can be formed print '-1'.

Input Description:The input consists of a single integer N, where 1 <= N <= 100000.

Output Description:Print the distinct pairs of prime numbers (prime x prime = N) in descending order. If no such pairs can be formed, print '-1'.

Sample Input:65

Sample Output:13 5

GOKUL (gokulmahendra246@gmail.com)

Completion Status: Completed

Concepts Included:

array

mathematics

Language Used: PYTHON 3

Source Code:

```
def is_prime(num):
if num < 2:
return False
if num == 2:
return True
if num % 2 == 0:
return False
for i in range(3, int(num**0.5) + 1, 2):
```

```
if num % i == 0:
return False
return True

def prime_pairs(N):
pairs = []
for p in range(2, int(N**0.5) + 1):
if N % p == 0:
q = N // p
if is_prime(p) and is_prime(q):
pairs.append((max(p, q), min(p, q)))
if pairs:
for pair in pairs:
print(*pair)
else:
print(-1)

# Driver code
N = int(input().strip())
prime_pairs(N)
```

GOKUL (gokulmahendran46@gmail.com)

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

13 3

Compilation Status: Passed

Execution Time:

0.01s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

-1

Compilation Status: Passed

Execution Time:

0.015s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

5 5

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

13 5

Compilation Status: Passed

Execution Time:

0.015s

GOKUL (gokulmahendran46@gmail.com)

101. Problem Statement:Given a number N, check if it is a power of 2.

Input Description:The input consists of a number N, where 1 <= N <= 100000.

Output Description:Print 'yes' if N is a power of 2, otherwise print 'no'.

Sample Input:64

Sample Output:yes

Completion Status: Completed

Concepts Included:

mathematics

Language Used: PYTHON 3

Source Code:

```
def is_power_of_two(n: int) -> str:
if n > 0 and (n & (n - 1)) == 0:
return "yes"
else:
return "no"

# Driver code
N = int(input().strip())
print(is_power_of_two(N))
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.009s

TestCase2:

Input:

< hidden >

Expected Output:

GOKUL (gokulmahendran46@gmail.com)

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.01s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.01s

TestCase4:

Input:

< hidden >

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.01s

TestCase5:

Input:

< hidden >

GOKUL (gokulmahendran46@gmail.com)

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.015s

102. Problem Statement:Given 2 numbers N and K.check if N is a power of K.Print 'yes' if it is a power of k otherwise print 'no'.

Sample Input:64 8

Sample Output:yes

Completion Status: Completed

Concepts Included:

mathematics

Language Used: PYTHON 3

Source Code:

```
def is_power(N, K):
if K == 1:
return "yes" if N == 1 else "no"
while N % K == 0:
N //= K
return "yes" if N == 1 else "no"

# Driver code
N, K = map(int, input().split())
print(is_power(N, K))
```

Compilation Details:

TestCase1:

Input:

< hidden >

Expected Output:

GOKUL (gokulmahendran46@gmail.com)

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.015s

TestCase2:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.015s

TestCase3:

Input:

< hidden >

Expected Output:

< hidden >

Output:

no

Compilation Status: Passed

Execution Time:

0.011s

TestCase4:

Input:

< hidden >

GOKUL (gokulmahendran46@gmail.com)

Expected Output:

< hidden >

Output:

yes

Compilation Status: Passed

Execution Time:

0.01s

GOKUL (gokulmahendran46@gmail.com)